

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch

```
bryan@WIN-R02UQQ0R3FN MINGW64 ~/source/repos/IT314AB_QUINO (feature/student-list)
$
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

- Your student list
- At least five student entries

```
class Student {
  final String? name;
  final String? course;
  final String? hobby;
  final String profileImage;
  final double? height;
  final bool? studentStatus;
  final int? age;

  const Student({
    required this.profileImage,
    this.name,
    this.course,
    this.hobby,
    this.height,
    this.studentStatus,
    this.age,
  });
}
```

```
final List<Student> students = [
  const Student(
    profileImage: 'lib/assets/lastikman.jpg',
    name: 'Bryan E. Quiño',
    course: '3 - BSIT',
    hobby: 'Basketball',
    height: 5.7,
    studentStatus: true,
    age: 21,
  ),

  const Student(
    profileImage: 'lib/assets/gagamboy.jpg',
    name: 'Dominic Ocarol',
    course: '3 - BSIT',
    height: 5.11,
    studentStatus: true,
    age: 22,
  ),

  const Student(
    profileImage: 'lib/assets/kapitan barbell.jpg',
    name: 'James Gulfan',
    hobby: 'Mobile Legends',
    height: 5.9,
    studentStatus: true,
    age: 20,
  ),

  const Student(
    profileImage: 'lib/assets/darna.jpg',
    name: 'Emji Syllanto',
    course: '3 - BSIT',
    hobby: 'Volleyball',
    height: 5.5,
    studentStatus: true,
    age: 23,
  ),

  const Student(
    profileImage: 'lib/assets/kidlat.jpg',
    name: 'Maverick Tajanlangit',
    course: '3 - BSME',
    height: 5.7,
    studentStatus: true,
    age: 22,
  ),
];
```

🚩 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

- Your five manually created student cards (do not change your past code yet)

```
const Text(  
  'Student Card 1',  
  style: TextStyle(  
    fontSize: 20,  
    fontWeight: FontWeight.bold,  
    color: Colors.black,  
    fontFamily: 'Times New Roman',  
  ), // TextStyle  
) // Text  
  
const SizedBox(height: 25),  
  
CircleAvatar(  
  radius: 50,  
  backgroundImage:  
    AssetImage(students[0].profileImage),  
) // CircleAvatar  
  
const SizedBox(height: 10),  
  
Text(  
  students[0].name ?? 'Not provided',  
  style: const TextStyle(  
    fontSize: 24,  
    fontWeight: FontWeight.bold,  
    color: Colors.deepPurple,  
    fontFamily: 'sans-serif',  
  ), // TextStyle  
) // Text  
  
const SizedBox(height: 10),  
  
Text(  
  'Course: ${students[0].course ?? 'Unknown'}',  
  style: const TextStyle(  
    fontSize: 20,  
    color: Colors.black87,  
    fontFamily: 'sans-serif',  
  ), // TextStyle  
) // Text  
  
const SizedBox(height: 10),  
  
Text(  
  'Hobby: ${students[0].hobby ?? 'Not provided'}',  
  style: const TextStyle(  
    fontSize: 20,  
    color: Colors.black87,  
    fontFamily: 'sans-serif',  
  ), // TextStyle  
) // Text  
  
const SizedBox(height: 10),  
  
Text(  
  'Student Status: '  
    '${students[0].studentStatus == null  
      ? 'Not provided' : studentStatus}',  
  style: const TextStyle(  
    fontSize: 16,  
    color: Colors.black87,  
    fontFamily: 'sans-serif',  
  ), // TextStyle  
) // Text
```

```
// Second Card  
Card(  
  elevation: 10,  
  shape: RoundedRectangleBorder(  
    borderRadius: BorderRadius.circular(10),  
  ), // RoundedRectangleBorder  
  
  child: Padding(  
    padding: const EdgeInsets.all(20),  
  
    child: Column(  
      mainAxisAlignment: MainAxisAlignment.min,  
      crossAxisAlignment: CrossAxisAlignment.center,  
  
      children: [  
  
        const Text(  
          'Student Card 2',  
          style: TextStyle(  
            fontSize: 20,  
            fontWeight: FontWeight.bold,  
            color: Colors.black,  
            fontFamily: 'Times New Roman',  
          ), // TextStyle  
) // Text  
  
        const SizedBox(height: 25),  
  
        CircleAvatar(  
          radius: 50,  
          backgroundImage:  
            AssetImage(students[1].profileImage),  
          ), // CircleAvatar  
  
        const SizedBox(height: 10),  
  
        Text(  
          students[1].name ?? 'Not provided',  
          style: const TextStyle(  
            fontSize: 24,  
            fontWeight: FontWeight.bold,  
            color: Colors.deepPurple,  
            fontFamily: 'sans-serif',  
          ), // TextStyle  
) // Text  
  
        const SizedBox(height: 10),  
  
        Text(  
          'Course: ${students[1].course ?? 'Unknown'}',  
          style: const TextStyle(  
            fontSize: 20,  
            color: Colors.black87,  
            fontFamily: 'sans-serif',  
          ), // TextStyle  
) // Text  
  
        const SizedBox(height: 10),  
  
        Text(  
          'Hobby: ${students[1].hobby ?? 'Not provided'}',  
          style: const TextStyle(  
            fontSize: 20,  
            color: Colors.black87,  
            fontFamily: 'sans-serif',  
          ), // TextStyle  
) // Text  
  
        const SizedBox(height: 10),  
  
        Text(  
          'Student Status: '  
            '${students[1].studentStatus == null  
              ? 'Not provided' : studentStatus}',  
          style: const TextStyle(  
            fontSize: 16,  
            color: Colors.black87,  
            fontFamily: 'sans-serif',  
          ), // TextStyle  
) // Text  
      ],  
    ),  
  ),  
)
```

```
// THIRD CARD  
Card(  
  elevation: 10,  
  shape: RoundedRectangleBorder(  
    borderRadius: BorderRadius.circular(10),  
  ), // RoundedRectangleBorder  
  
  child: Padding(  
    padding: const EdgeInsets.all(20),  
  
    child: Column(  
      mainAxisAlignment: MainAxisAlignment.min,  
      crossAxisAlignment: CrossAxisAlignment.center,  
  
      children: [  
  
        const Text(  
          'Student Card 3',  
          style: TextStyle(  
            fontSize: 20,  
            fontWeight: FontWeight.bold,  
            color: Colors.black,  
            fontFamily: 'Times New Roman',  
          ), // TextStyle  
) // Text  
  
        const SizedBox(height: 25),  
  
        CircleAvatar(  
          radius: 50,  
          backgroundImage:  
            AssetImage(students[2].profileImage),  
          ), // CircleAvatar  
  
        const SizedBox(height: 10),  
  
        Text(  
          students[2].name ?? 'Not provided',  
          style: const TextStyle(  
            fontSize: 24,  
            fontWeight: FontWeight.bold,  
            color: Colors.deepPurple,  
            fontFamily: 'sans-serif',  
          ), // TextStyle  
) // Text  
  
        const SizedBox(height: 10),  
  
        Text(  
          'Course: ${students[2].course ?? 'Unknown'}',  
          style: const TextStyle(  
            fontSize: 20,  
            color: Colors.black87,  
            fontFamily: 'sans-serif',  
          ), // TextStyle  
) // Text  
  
        const SizedBox(height: 10),  
  
        Text(  
          'Hobby: ${students[2].hobby ?? 'Not provided'}',  
          style: const TextStyle(  
            fontSize: 20,  
            color: Colors.black87,  
            fontFamily: 'sans-serif',  
          ), // TextStyle  
) // Text  
  
        const SizedBox(height: 10),  
  
        Text(  
          'Student Status: '  
            '${students[2].studentStatus == null  
              ? 'Not provided' : studentStatus}',  
          style: const TextStyle(  
            fontSize: 16,  
            color: Colors.black87,  
            fontFamily: 'sans-serif',  
          ), // TextStyle  
) // Text  
      ],  
    ),  
  ),  
)
```

```
// FOURTH CARD
Card(
  elevation: 10,
  shape: RoundedRectangleBorder(
    borderRadius: BorderRadius.circular(10),
  ), // RoundedRectangleBorder

  child: Padding(
    padding: const EdgeInsets.all(20),

    child: Column(
      mainAxisAlignment: MainAxisAlignment.min,
      crossAxisAlignment: CrossAxisAlignment.center,

      children: [

        const Text(
          'Student Card 4',
          style: TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
            color: Colors.black,
            fontFamily: 'Times New Roman',
          ), // TextStyle
        ), // Text

        const SizedBox(height: 25),

        CircleAvatar(
          radius: 50,
          backgroundImage:
            AssetImage(students[3].profileImage),
        ), // CircleAvatar

        const SizedBox(height: 10),

        Text(
          students[3].name ?? 'Not provided',
          style: const TextStyle(
            fontSize: 24,
            fontWeight: FontWeight.bold,
            color: Colors.deepPurple,
            fontFamily: 'sans-serif',
          ), // TextStyle
        ), // Text

        const SizedBox(height: 10),

        Text(
          'Course: ${students[3].course ?? 'Unknown'}',
          style: const TextStyle(
            fontSize: 20,
            color: Colors.black87,
            fontFamily: 'sans-serif',
          ), // TextStyle
        ), // Text

        const SizedBox(height: 10),

        Text(
          'Hobby: ${students[3].hobby ?? 'Not provided'}',
          style: const TextStyle(
            fontSize: 20,
```

```
// FIFTH CARD
Card(
  elevation: 10,
  shape: RoundedRectangleBorder(
    borderRadius: BorderRadius.circular(10),
  ), // RoundedRectangleBorder

  child: Padding(
    padding: const EdgeInsets.all(20),

    child: Column(
      mainAxisAlignment: MainAxisAlignment.min,
      crossAxisAlignment: CrossAxisAlignment.center,

      children: [

        const Text(
          'Student Card 5',
          style: TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
            color: Colors.black,
            fontFamily: 'Times New Roman',
          ), // TextStyle
        ), // Text

        const SizedBox(height: 25),

        CircleAvatar(
          radius: 50,
          backgroundImage:
            AssetImage(students[4].profileImage),
        ), // CircleAvatar

        const SizedBox(height: 10),

        Text(
          students[4].name ?? 'Not provided',
          style: const TextStyle(
            fontSize: 24,
            fontWeight: FontWeight.bold,
            color: Colors.deepPurple,
            fontFamily: 'sans-serif',
          ), // TextStyle
        ), // Text

        const SizedBox(height: 10),

        Text(
          'Course: ${students[4].course ?? 'Unknown'}',
          style: const TextStyle(
            fontSize: 20,
            color: Colors.black87,
            fontFamily: 'sans-serif',
          ), // TextStyle
        ), // Text

        const SizedBox(height: 10),
```

🚩 Flag 3 — Discover **ListView**

Objective: Make your student directory scrollable using **ListView**.

Research Flutter's **ListView**.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

Test

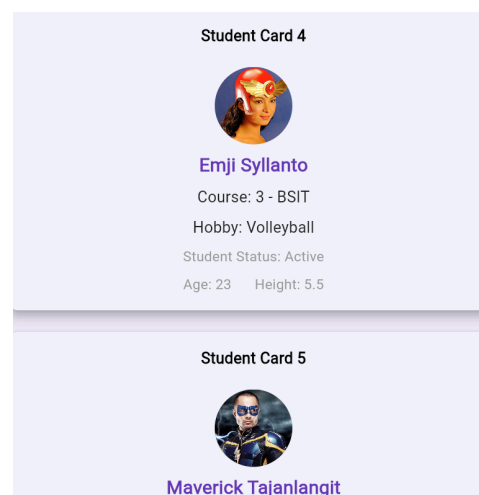
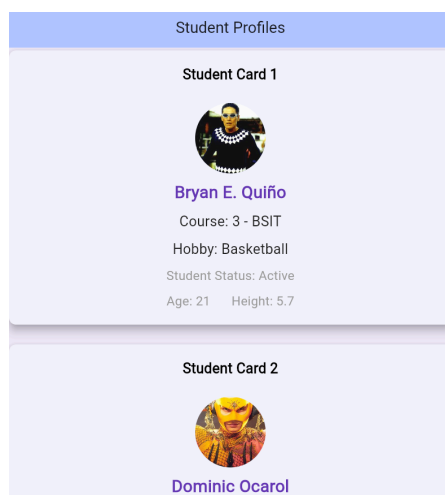
The user should be able to:

- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- **ListView** in your code
- Top of the application
- Bottom portion showing later students

```
body: ListView(  
  children: [  
    Card(  
      elevation: 10,  
      shape: RoundedRectangleBorder(  
        borderRadius: BorderRadius.circular(10),  
      ), // RoundedRectangleBorder  
  
      child: Padding(  
        padding: const EdgeInsets.all(20),  
  
        child: Column(  
          mainAxisAlignment: MainAxisAlignment.min,  
          crossAxisAlignment: CrossAxisAlignment.center,  
  
          children: [  
            const Text(  
              'Student Card 1',  
            ),  
            const Text(  
              'Student Card 2',  
            ),  
            const Text(  
              'Student Card 3',  
            ),  
            const Text(  
              'Student Card 4',  
            ),  
            const Text(  
              'Student Card 5',  
            ),  
          ],  
        ),  
      ),  
  ],  
)
```



🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

Without creating another card manually, your application should display the new student automatically.

Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app

```
body: ListView.builder(  
  padding: const EdgeInsets.all(20),  
  itemCount: students.length,  
  itemBuilder: (context, index) {  
    final student = students[index];  
    return Card(  
      elevation: 10,  
      shape: RoundedRectangleBorder(  
        borderRadius: BorderRadius.circular(10),  
      ), // RoundedRectangleBorder  
  
      child: Padding(  
        padding: const EdgeInsets.all(20),  
  
        child: Column(  
          mainAxisAlignment: MainAxisAlignment.min,  
          crossAxisAlignment: CrossAxisAlignment.center,  
  
          children: [  

```

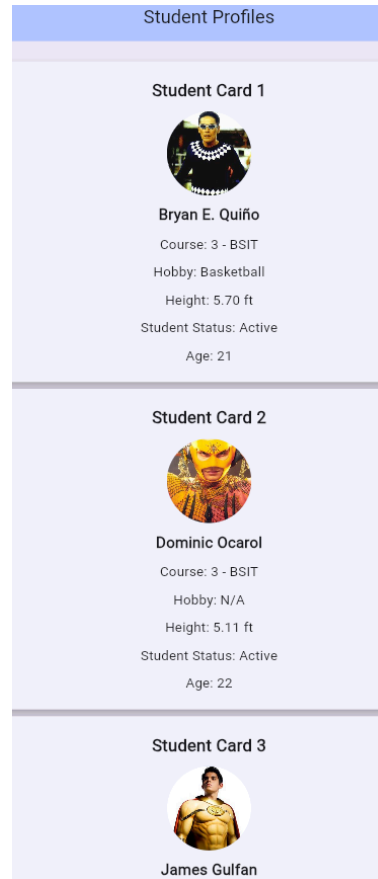
```

class Student {
    final String? name;
    final String? course;
    final String? hobby;
    final String? profileImage;
    final double? height;
    final bool? studentStatus;
    final int? age;

    const Student({
        required this.profileImage,
        this.name,
        this.course,
        this.hobby,
        this.height,
        this.studentStatus,
        this.age,
    });
}

final List<Student> students = [
    const Student(
        profileImage: 'lib/assets/lastikman.jpg',
        name: 'Bryan E. Quiño',
        course: '3 - BSIT',
        hobby: 'Basketball',
        height: 5.7,
        studentStatus: true,
        age: 21,
    ),
    const Student(
        profileImage: 'lib/assets/gagamboy.jpg',
        name: 'Dominic Ocarol',
        course: '3 - BSIT',
        height: 5.11,
        studentStatus: true,
        age: 22,
    ),
]

```



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields

```
final List<Student> students = [
  const Student(
    profileImage: 'lib/assets/lastikman.jpg',
    name: 'Bryan E. Quiño',
    course: '3 - BSIT',
    hobby: 'Basketball',
    height: 5.7,
    studentStatus: true,
    age: 21,
    studentID: 'S001',
    email: 'bryan.quino@dtbc-cebu.edu.ph',
    favoriteSubject: 'Computer Science',
  ),
  const Student(
    profileImage: 'lib/assets/gagamboy.jpg',
    name: 'Dominic Ocarol',
    course: '3 - BSIT',
    hobby: 'Video Games',
    height: 5.11,
    studentStatus: true,
    age: 22,
    studentID: 'S002',
    email: 'dominic.ocarol@dtbc-cebu.edu.ph',
    favoriteSubject: 'Mathematics',
  ),
  const Student(
    profileImage: 'lib/assets/kapitan barbell.jpg',
    name: 'James Gulfan',
    course: '3 - BSIT',
    hobby: 'Mobile Legends',
    height: 5.9,
    studentStatus: true,
    age: 20,
    studentID: 'S003',
    email: 'james.gulfan@dtbc-cebu.edu.ph',
    favoriteSubject: 'Physics',
  ),
];
```

```
const Student(
  profileImage: 'lib/assets/darna.jpg',
  name: 'Emji Syllanto',
  course: '3 - BSIT',
  hobby: 'Volleyball',
  height: 5.5,
  studentStatus: true,
  age: 23,
  studentID: 'S004',
  email: 'emji.syllanto@dtbc-cebu.edu.ph',
  favoriteSubject: 'Biology',
),
const Student(
  profileImage: 'lib/assets/kidlat.jpg',
  name: 'Maverick Tajanlangit',
  course: '3 - BSIT',
  hobby: 'Dancing',
  height: 5.7,
  studentStatus: true,
  age: 22,
  studentID: 'S005',
  email: 'maverick.tajanlangit@dtbc-cebu.edu.ph',
  favoriteSubject: 'Chemistry',
),
];
```

Student Card 1



ID: S001

Bryan E. Quiño

Email: bryan.quino@dtbc-cebu.edu.ph

Course: 3 - BSIT

Age: 21

Height: 5.70 ft

Hobby: Basketball

Favorite Subject: Computer Science

Student Status: Active

Student Card 2



ID: S002

Dominic Ocarol

Email: dominic.ocarol@dtbc-cebu.edu.ph

Course: 3 - BSIT

Age: 22

Height: 5.11 ft

Hobby: Video Games

Favorite Subject: Mathematics

Student Status: Active

🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

```
[]
```

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

- "No students found."
- "Student list is empty."

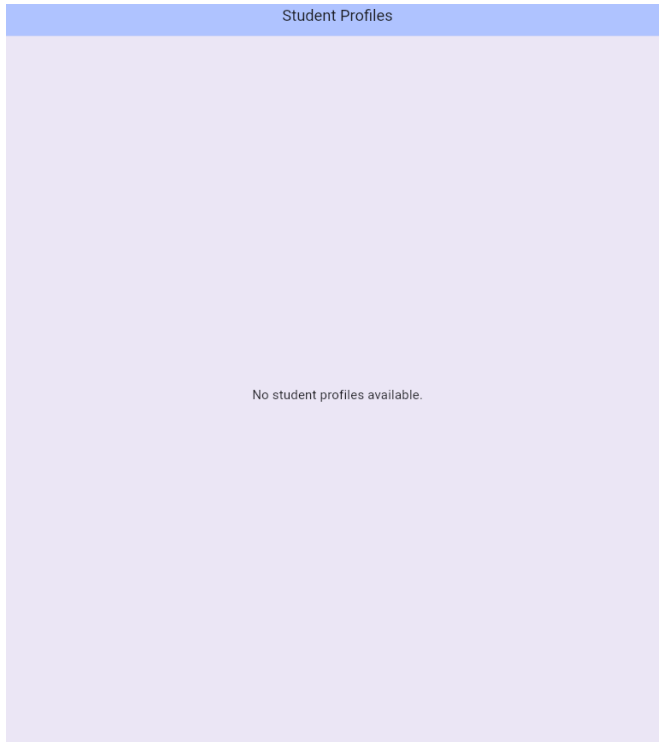
Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

- Empty student list
- Empty-state UI
- Application still functioning correctly

```
final List<Student> students = [];
```



Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a [List](#).

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

After sorting:

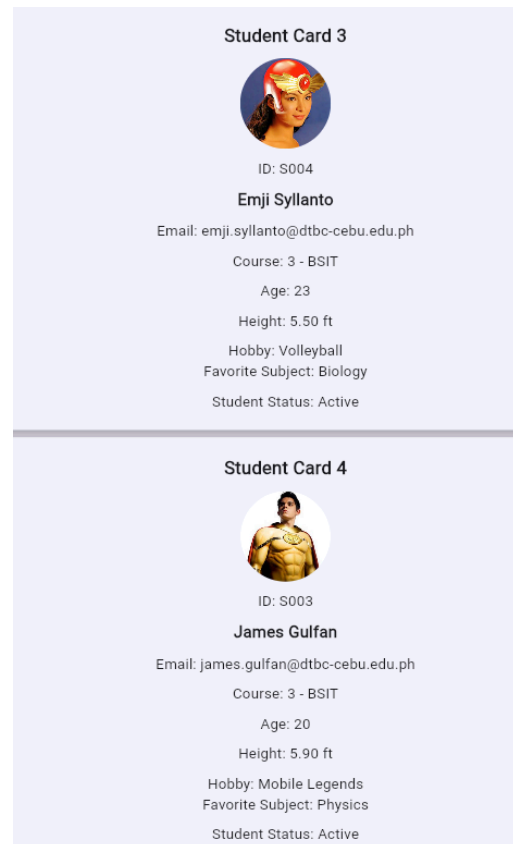
- Aaron
- John
- Maria
- Sophia

Your UI should immediately reflect the new order.

Screenshot:

- Student list before or after sorting
- Application displaying the sorted order

```
const Student(  
  profileImage: 'lib/assets/kapitan barbell.jpg',  
  name: 'James Gulfan',  
  course: '3 - BSIT',  
  hobby: 'Mobile Legends',  
  height: 5.9,  
  studentStatus: true,  
  age: 20,  
  studentID: 'S003',  
  email: 'james.gulfan@dtbc-cebu.edu.ph',  
  favoriteSubject: 'Physics',  
),  
  
const Student(  
  profileImage: 'lib/assets/darna.jpg',  
  name: 'Emji Syllanto',  
  course: '3 - BSIT',  
  hobby: 'Volleyball',  
  height: 5.5,  
  studentStatus: true,  
  age: 23,  
  studentID: 'S004',  
  email: 'emji.syllanto@dtbc-cebu.edu.ph',  
  favoriteSubject: 'Biology',  
),
```



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

feature/student-list

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
bryan@WIN-RO2UQQ0R3FN MINGW64 ~/source/repos/IT314AB_QUINO (feature/student-list)
$ git commit -m "feat: added capture the flag 4 requirements"
[feature/student-list 518eb5d] feat: added capture the flag 4 requirements
1 file changed, 75 insertions(+), 534 deletions(-)
```

```
bryan@WIN-RO2UQQ0R3FN MINGW64 ~/source/repos/IT314AB_QUINO (feature/student-list)
$ git push origin feature/student-list
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 1.33 KiB | 1.33 MiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote:
remote: Create a pull request for 'feature/student-list' on GitHub by visiting:
remote:   https://github.com/bryankinyo/IT314AB_QUINO/pull/new/feature/student-list
remote:
To https://github.com/bryankinyo/IT314AB_QUINO.git
 * [new branch]      feature/student-list -> feature/student-list
```

 **IT314AB_QUINO** Public Pin Watch

 feature/student-list had recent pushes 37 seconds ago Compare & pull request

 feature/student-list  5 Branches  0 Tags Add file Code

This branch is 9 commits ahead of `main`. Contribute

 **bryankinyo**

 feat: added capture the flag 4 requirements 518eb5d · 1 minute ago 11 Commits

 my_first_flutter_app_kinyo	feat: added capture the flag 4 requirements	1 minute ago
 IT314AB - Capture The Flag 2.pdf	docs: added capture the flag v2 pdf	last week
 IT314AB - Capture The Flag_FlutterActivity.pdf	docs: added capture the flag flutter activity	2 weeks ago
 IT314AB Capture The Flag 3.pdf	docs: added capture the flag 3 pdf	5 days ago
 IT314AB_QUINO.pdf	docs: add assignment pdf	3 weeks ago
 README.md	Initial commit	3 weeks ago